

Tape Emulator — User Manual

Saving and loading from BASIC, using a virtual cassette held in a single `.tap` file.

CHEAT SHEET

TO DO THIS	TYPE / PRESS
Save the current program	<code>S</code> <code>R.Shift</code> + <code>P</code> <i>name</i> <code>R.Shift</code> + <code>P</code> <code>Enter</code> , then any key at the prompt
Load the next program	<code>J</code> <code>R.Shift</code> + <code>P</code> <code>R.Shift</code> + <code>P</code> <code>Enter</code>
Load a program by name	<code>J</code> <code>R.Shift</code> + <code>P</code> <i>name</i> <code>R.Shift</code> + <code>P</code> <code>Enter</code>
Check a save against memory	<code>L.Shift</code> + <code>R.Shift</code> , <code>R.Shift</code> + <code>R</code> , <code>R.Shift</code> + <code>P</code> <i>name</i> <code>R.Shift</code> + <code>P</code> <code>Enter</code>
Rewind the tape	<code>F5</code>
Re-read the file from disk	<code>F6</code>
Start a fresh, blank tape	<code>F7</code>
Escape a stuck load (BREAK)	<code>L.Shift</code> + <code>Space</code>

THE MODEL

The emulator has one virtual cassette in its deck, stored as `tape.tap` in the directory you launched it from. If that file is not there, the emulator starts anyway and the first **SAVE** creates it.

It behaves like a real cassette rather than a folder of files:

- Recording always goes on the **end**. Saving never overwrites what is already on the tape.
- Loading searches **forward only** from wherever the head is sitting. It never rewinds by itself.
- The tape has one head position, shared by loading and saving.

The whole tape is written to disk after every block, so an emulator that is killed mid-session still leaves a usable file.

SAVING

Type **SAVE "name"** and press `Enter`. The ROM answers with `Start tape`, then press **any key**. — press any key and the save completes instantly.

YOU WANT	PRESS	NOTES
SAVE	<code>S</code>	A keyword, so one key at the start of a line
<code>"</code>	<code>R.Shift</code> + <code>P</code>	Right Shift is SYM SHIFT
the name	ordinary letters	Up to 10 characters; longer names are truncated
CODE	<code>L.Shift</code> + <code>R.Shift</code> , then <code>I</code>	Both Shifts together is EXTENDED mode

```
Save a BASIC program
SAVE "myprog"

Save a block of memory: 6912 bytes from the screen
SAVE "screen"CODE 16384,6912
```

Repeated saves accumulate. Saving the same program twenty times leaves twenty copies on the tape, and a load by name finds the *oldest* one. Press `F7` before a session, or delete `tape.tap`, to start clean.

Loading & Tape Controls

Reading files back, and the three function keys that drive the deck.

LOADING

Type **LOAD ""** and press **Enter**. There is no prompt — loading starts immediately. As the ROM walks the tape it announces each file it passes, which is how you can see where it got to.

```
Load the next program on the tape
LOAD ""

Find a particular one, skipping anything before it
LOAD "myprog"

Load a memory block back to the address it was saved from
LOAD ""CODE
```

Press F5 before each load. A successful load leaves the head just past that file, so loading two files in tape order works, but going back to an earlier one does not. One rewind reaches anything: the ROM walks forward past every file it is not looking for.

FUNCTION KEYS

The Spectrum keyboard uses every key it has, so the tape controls live on function keys, which the emulated matrix does not map. Each prints a line to the terminal confirming what it did.

KEY	DOES	DETAIL
F5	Rewind	Moves the head back to the start of the tape. Nothing is written or lost.
F6	Reload	Re-reads tape.tap from disk and rewinds. Use it if you swapped the file from outside, or to undo an F7.
F7	New tape	Starts a blank tape. The file on disk is left alone until your next save rewrites it, so this is safe to press by accident — F6 brings the old tape straight back.

A WORKED SESSION

```
F7      start a fresh tape
SAVE "p"  tape.tap now holds  p
SAVE "q"  tape.tap now holds  p q
SAVE "r"  tape.tap now holds  p q r

F5      rewind
LOAD "q"  walks past p (announcing it), loads q

F5      rewind again -- p is behind the head now
LOAD "p"  loads p
```

Behaviour & Troubleshooting

Exactly what each operation does to the tape, and what to do when something looks wrong.

WHAT THE DECK DOES, PRECISELY

ACTION	EFFECT ON THE TAPE
SAVE	Appends a header block and a data block to the end. Does not move the head , so on a fresh tape you can save and then load straight back with no rewind.
LOAD / VERIFY	Reads forward from the head, skipping files that do not match, and leaves the head just past the file it loaded.
Reaching the end	Loading reports failure. The ROM keeps asking, so it will appear to hang — that is genuine Spectrum behaviour for a tape that has run out.
F7 then SAVE	The file is rewritten containing only what you save from then on.

If a load seems to hang, press **L.Shift** + **Space** — that is BREAK on a Spectrum. The usual causes are a name that is not on the tape, or a head sitting past the file you asked for (press **F5** and try again).

TROUBLESHOOTING

SYMPTOM	CAUSE AND FIX
Load never finishes	Name not on the tape, or the head is past it. BREAK out with L.Shift + Space , press F5 , check the spelling.
Loaded an old version	Repeated saves accumulate and the search finds the oldest match. F7 and save once.
Tape 'tape.tap': not loaded at startup	Normal when no tape exists yet. The first SAVE creates it.
A save appears to do nothing	Check the terminal — a write failure is reported there. SAVE cannot report errors to the Spectrum, because a real recorder cannot answer back.
A game will not load	Very likely a turbo loader — see overleaf.

NOTES FOR THE CURIOUS

- Loading into `0x0000–0x3FFF` silently does nothing. That region is ROM, and the hardware ignores writes there.
- VERIFY** works. Press **L.Shift** + **R.Shift**, then **R.Shift** + **R**, then the name. It compares rather than stores; a mismatch reports `R Tape loading error`.
- The tape is only ever touched by the CPU thread, so the function keys set a request that is picked up at the next frame boundary.

Compatibility & Internals

What the emulator will and will not load, and how the tape is actually wired to the ROM.

WHAT WORKS

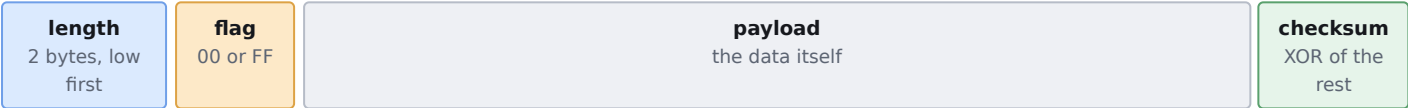
- **Any standard-speed `.tap` file**, from anywhere. Files with several programs on them load correctly, and **LOAD "name"** picks one out of the middle.
- **Files this emulator writes are ordinary `.tap` files** — other emulators read them, and vice versa.
- **Damaged tapes still load as far as they go.** A bad checksum is reported rather than refused, and a truncated file is used up to the point of damage.

Turbo and custom loaders do not work. Many commercial games load a small stub through the ROM and then take over, reading the tape signal themselves. This emulator intercepts the ROM's own tape routines, so a loader that bypasses them gets no data and stalls. Standard-loader titles and type-ins are fine.

`.ttx` files are not supported at all — that format exists largely to describe the turbo timings above.

THE `.TAP` FORMAT

A `.tap` file is nothing but blocks laid end to end. There is no file header, no index and no end marker, which is why an empty file is a valid blank tape.



`length` counts the flag and checksum as well as the payload. `flag` is `00` for a header and `FF` for data. Every saved file is a header block immediately followed by its data block.

Inside a header block

OFFSET	SIZE	MEANING
0	1	Type: 0 Program, 1 Number array, 2 Character array, 3 Code
1–10	10	Filename, padded with spaces
11–12	2	Length of the data block that follows
13–14	2	Program: autostart line (≥ 32768 means none). Code: start address
15–16	2	Program: offset to the variables area. Code: unused

HOW IT IS WIRED

The emulator does not pretend to be a tape recorder producing an audio signal. Instead it intercepts the two ROM routines that would have listened to one, and moves the bytes directly:

ADDRESS	ROM ROUTINE	REPLACED BY
0x0556	LD-BYTES	Reads the next block off the virtual tape into memory
0x04C2	SA-BYTES	Appends a block to the virtual tape and writes the file

That is why loading is instant rather than taking the couple of minutes a real cassette would, and why the border does not stripe during a load. It is also exactly why turbo loaders fail: they never call these routines.

The interception steps aside when there is no tape in the deck, or when BREAK is held. In both cases the genuine ROM code runs instead, so the machine behaves as it would with an empty recorder — including being able to abort cleanly.

ZX Spectrum emulator — tape module. Behaviour described here is exercised by the project's automated test suite; the ROM routine addresses above were read out of `48.rom` rather than taken from documentation. See `docs/keyboard-map.pdf` for the full Spectrum-to-PC key mapping.